

PMPML AI-CONNECT

Smart Transit Ecosystem & Intelligent Fleet Management

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APCOER

1. SYSTEM ARCHITECTURE

A multi-layered transit prototype designed to eliminate "Ghost Bus" uncertainty through real-time coordinate simulation and local AI inference.

Layer	Technology & Function	
Frontend UI	Leaflet.js OSRM	Real-time mapping of 5 primary Pune bus routes (149, 174, 197, 201, 204) following actual road geometries.
Backend Logic	Flask Python	Orchestrates data flow, manages traffic congestion zones, and computes efficiency scores for every active bus.
AI Engine	Ollama Llama 3	Local LLM integration for contextual travel advice. 100% offline, private, and zero-latency.

2. DYNAMIC EFFICIENCY SCORING

The core intelligence relies on a weighted algorithm that calculates the optimal travel option for students heading to Keystone School of Engineering.

$$\text{Score} = (P \times 0.5) - (D \times 0.3) - (T \times 0.2)$$

P: Progress towards destination | **D:** Crowd Density Factor | **T:** Traffic Delay Penalty

- Predictive Routing:** The system prioritizes buses based on overall efficiency rather than simple proximity.
- Adaptive Learning:** AI adjusts recommendations in real-time as buses enter traffic-heavy zones like Swargate or Hadapsar.

3. INTELLIGENT FLEET FEATURES

Traffic Simulation

Dynamic speed reduction based on defined congestion zones (Moderate to Heavy delay).

Stop-Level Query

Interactive stops allow users to query Llama 3 for precise arrival estimates at any specific point on the route.

Presence Verification

Crowd-sourced trust layer matching user GPS with bus coordinates to validate physical location.

4. STRATEGIC ADVANTAGE

Unlike global mapping solutions, PMPML AI-Connect is hyper-localized for the Pune transit ecosystem:

- **Zero-Cost Infrastructure:** Utilizing open-weight models (Llama 3) via Ollama eliminates cloud API subscriptions.
- **Data Privacy:** Student travel patterns remain local and encrypted within the system's edge environment.
- **Offline Resilience:** The entire system operates without an internet connection, essential for high-reliability transit tracking.